

TECHNICAL CASE STUDY

PII Airlock

A local privacy gateway for cloud AI requests

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Status	Reproducible public demonstrator

Outcome first

A working two-contour gateway now turns a document and task into a pseudonymized cloud payload, blocks unsafe cases, and restores only exact operation-scoped tokens. Cloud use is optional; the default is an inspectable dry-run. The project does not claim perfect anonymization.

1. Problem and design decision

A personal AI assistant can benefit from a cloud model while its local context contains names, contacts, addresses, IDs, and credentials. A local LLM alone is not a security boundary: it can miss an entity, invent a substring, return invalid JSON, or follow an instruction embedded in the document. The design therefore combines a semantic detector with deterministic checks.

2. Two-contour flow

- LM Studio receives the task and document only through 127.0.0.1 and returns schema-constrained entity candidates.
- Code accepts exact source substrings, resolves overlap longest-first, and emits `__PII_<nonce>_<TYPE>_<N>__` tokens.
- A residual gate checks the exact outgoing fields. Any model/schema error or known residual secret stops the request.
- OpenAI is optional. Without a key the exact payload is shown; with a key the Responses API is called with `store: false`.
- The cloud response is untrusted. Unknown, altered, or forged tokens block restoration; the in-memory map is destroyed.

LOCAL → CLOUD → LOCAL

PII Airlock

A local privacy boundary for cloud LLM requests. Raw values stay on this machine.

LM Studio: reachable · Cloud: dry-run

LOCAL DETECTOR
Qwen 3.5 9B

TASK
Write a concise, polite reply while preserving every PII token exactly.

LOAD TXT / MD / DOCX / PDF
Выберите файл | Файл не выбран

01 · LOCAL INPUT
Original document

Contact Elena Morozova at elena.morozova@example.test or +7 999 123-45-67. Contract CN-2049 belongs to Northstar Lab.

02 · LOCAL DETECTION
Detected types

CONTRACT_ID × 1
EMAIL × 1
ORG × 1
PERSON × 1
PHONE × 1

03 · CLOUD PAYLOAD
Pseudonymized text

Contact __PII_F8938AF4_PERSON_0001__ at __PII_F8938AF4_EMAIL_0001__ or __PII_F8938AF4_PHONE_0001__. Contract __PII_F8938AF4_CONTRACT_ID_0001__ belongs to __PII_F8938AF4_ORG_0001__.

04 · LOCAL RESTORE
Final answer

DRY RUN — sanitized payload was not sent because OPENAI_API_KEY is absent.

SAFE · DRY RUN
Analyze locally
Send sanitized payload
Destroy mapping

Real local Qwen dry-run · synthetic values only

3. Evidence, not promises

All figures below come from the new reproducible run on 30 synthetic documents (15 Russian, 15 English).

Metric	Qwen 3.5 9B	Gemma 4 E4B
Approved payloads / 30	25	25
Control leaks in approved payloads	0	0
Entity-level recall	0.6667	0.7593
Extra replacement values	5	4
Schema / exact-substring failures	4	0
Dataset release blocks	1	5
Round-trip fidelity cases	26	30
Median / maximum	2.556 / 6.182 s	1.567 / 4.116 s

Interpretation: both models remain experimental. The dataset-only oracle withheld any payload still containing a known control value. Zero approved-payload leaks is bounded evidence for these fixtures—not proof for unseen documents.

4. Delivered artifacts

- Python 3.11 Web UI, CLI, and local API with a 10-minute in-memory mapping TTL.
- TXT, Markdown, DOCX, and text-layer PDF parsing; OCR/images/archives and >20,000 characters blocked in V1.
- 25 offline tests with stubs: span validation, overlap, tokens, TTL, formats, timeout, prompt-injection framing, logs, and round trip.
- Full English/Russian README, threat model, SECURITY.md, synthetic fixtures, per-case JSON, screenshot, and GitHub Actions.
- A local, disabled Gosha adapter with fail-closed behavior; 27/27 current Gosha tests passed and no service was restarted.

5. Risks and next experiment

Semantic PII can still be missed; rules are not universal DLP. Process memory, LM Studio internals, OS swap, and provider policy are outside the guarantee. High-risk use requires dry-run review. The next useful experiment is a larger adversarial fixture set plus improved semantic detection without weakening exact-substring validation.